

MTE 204 Problem Set #4: Solutions

Regression and Interpolation (Chapter 17 and 18 in Chapra 8th edition, International Version)

17.3 Use least-squares regression to fit a straight line to

x	0	2	4	6	9	11	12	15	17	19
y	4	6	7	6	9	8	8	10	12	12

Along with the slope and intercept, compute the standard error of the estimate and the correlation coefficient. Plot the data and the regression line. Then repeat the problem, but regress x versus y —that is, switch the variables. Interpret your results.

The results can be summarized as

	y versus x	x versus y
Best fit equation	$y = 4.5471295 + 0.3845127x$	$x = -9.6688312 + 2.3376623y$
Standard error	2.616188916	6.450667493
Correlation coefficient (r)	0.9481	0.9481
Coefficient of Determination (r^2)	0.898860819	0.898860819

We can solve the second fit for the line on the same axes:

$$x = a_0 + a_1 y$$

$$y = \frac{x}{a_1} - \frac{a_0}{a_1} = \frac{1}{2.337662}x - \frac{-9.66883}{2.337662} = 0.427778x + 4.13611$$

We can now plot both lines on the same graph

We did not cover standard error in class

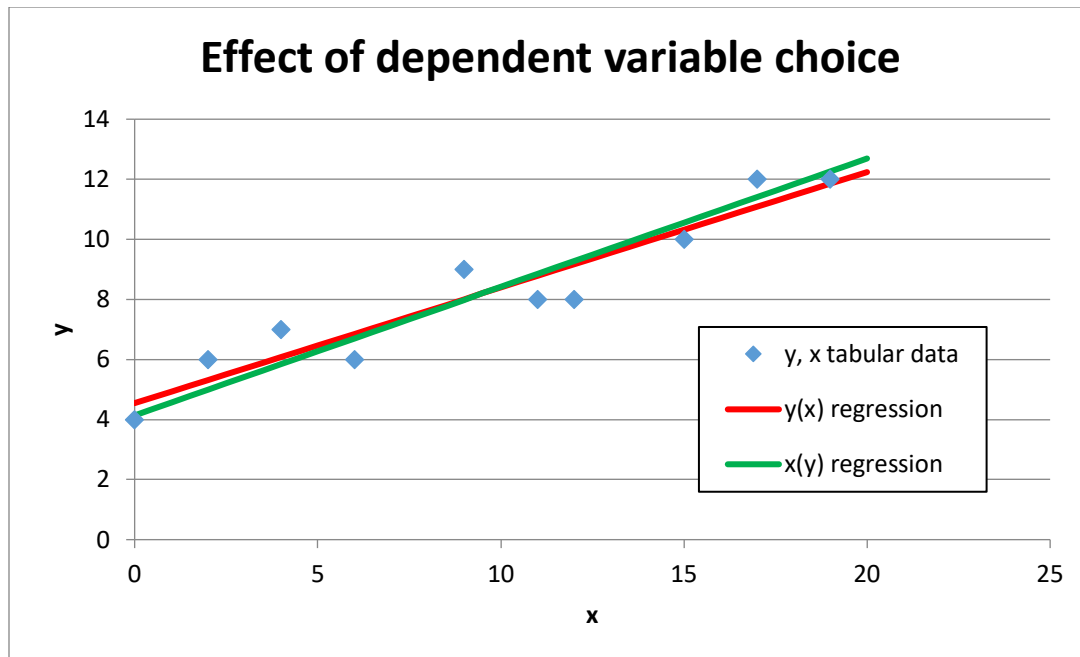
$$s_{y/x} = \sqrt{\frac{S_r}{n-2}}$$

where $s_{y/x}$ is called the **standard error** of the estimate.

The value provided in the solution is not the standard error of the estimate, but rather the standard deviation s_y , which is defined below.

$$s_y = \sqrt{\frac{S_t}{n-1}}$$

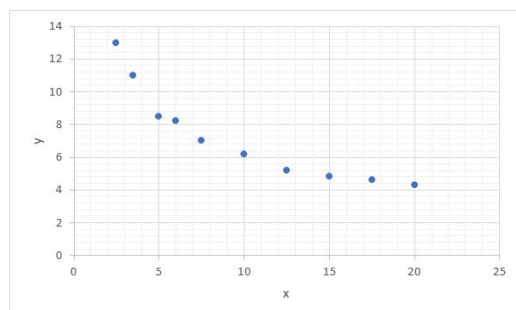
For this problem, the standard error of the estimate $s_{y/x}$ for y vs x is 0.8825. The standard deviation s_y is 2.6162.



The “best” fit lines and standard errors differ depending on the choice of independent and dependent variable. This makes sense because in the development of the technique, the independent variable is assumed to have negligible error. We thus minimize the residuals with different assumptions, leading to slightly different values for the fit parameters. The standard error calculation is significantly different between the two choices, with the y as dependent having much smaller standard error than x as dependent. This makes sense if x is actually an independent rather than stochastic variable. In contrast, the correlation coefficients are identical since this statistic is measuring how well the linear regression method models the set of points, regardless of dependent / independent choice. This can also be thought of as the same amount of uncertainty is explained through linear regression, regardless of how the data points are plotted.

17.8 Fit the following data with the power model ($y = ax^b$). Use the resulting power equation to predict y at $x = 9$:

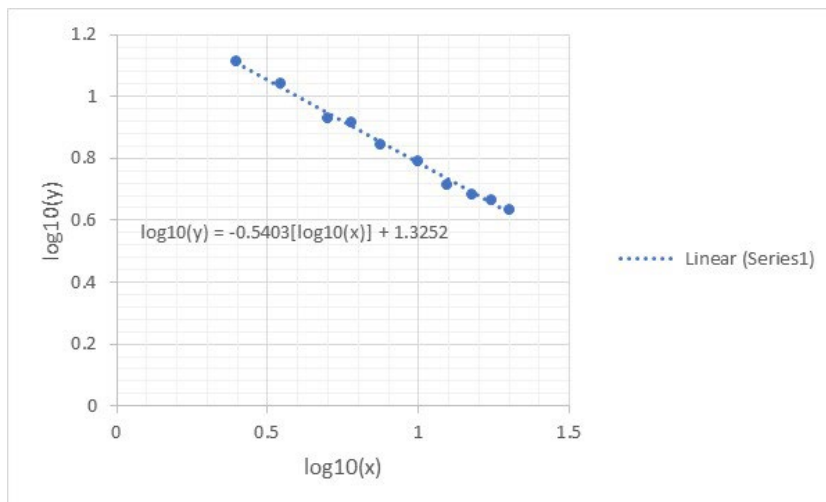
x	2.5	3.5	5	6	7.5	10	12.5	15	17.5	20
y	13	11	8.5	8.2	7	6.2	5.2	4.8	4.6	4.3



$\log_{10}(x)$	$\log_{10}(y)$
0.39794	1.113943
0.544068	1.041393
0.69897	0.929419
0.778151	0.913814
0.875061	0.845098
1	0.792392
1.09691	0.716003
1.176091	0.681241
1.243038	0.662758
1.30103	0.633468

We regress $\log_{10}(y)$ versus $\log_{10}(x)$ to give

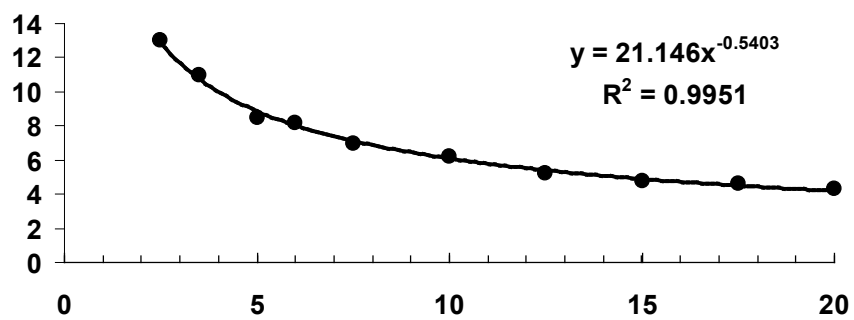
$$\log_{10} y = 1.325225 - 0.54029 \log_{10} x$$



Therefore, $\alpha_2 = 10^{1.325225} = 21.14583$ and $\beta_2 = -0.54029$, and the power model is

$$y = 21.14583x^{-0.54029}$$

The model and the data can be plotted as



The model can be used to predict a value of $21.14583(9)^{-0.54029} = 6.451453$.

17.9 Fit an exponential model to

x	0.4	0.8	1.2	1.6	2	2.3
y	800	980	1500	1945	2900	3600

Plot the data and the equation on both standard and semilogarithmic graph paper.

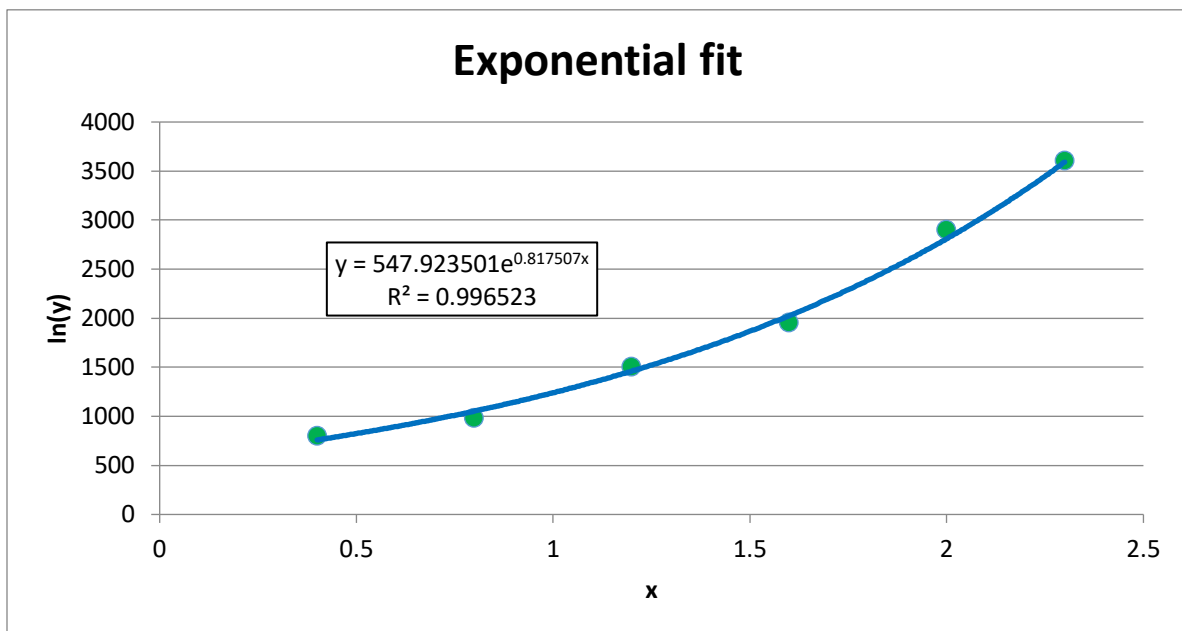
We regress $\ln(y)$ versus x to give

$$\ln y = 6.306136 + 0.817507x$$

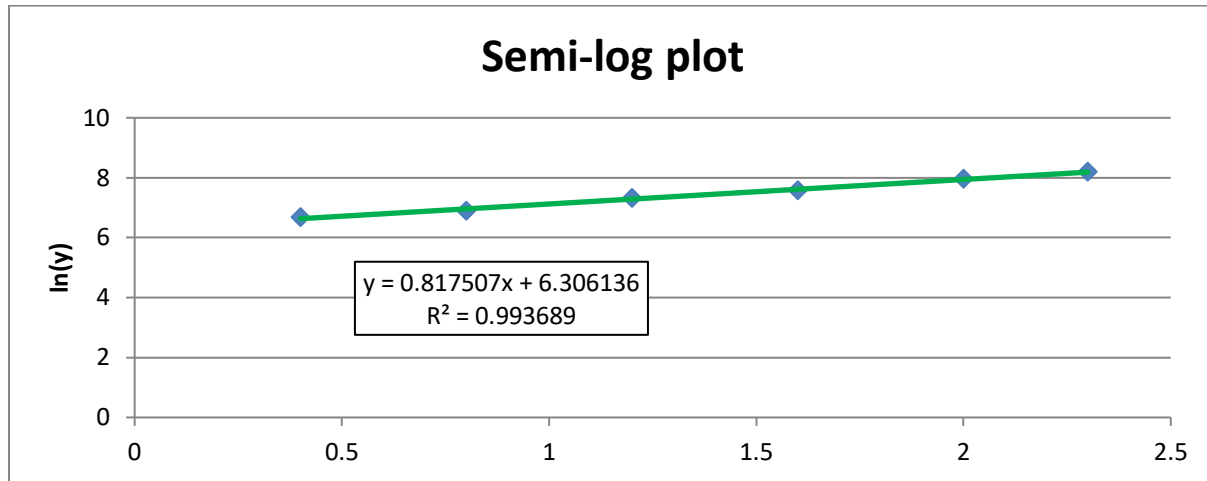
Therefore, $\alpha_1 = e^{6.306136} = 547.923501$ and $\beta_1 = 0.817507$, and the exponential model is

$$y = 547.923501e^{0.817507x}$$

The model and the data can be plotted as



A semi-log plot can be developed by plotting the natural log y versus x . As expected, both the data and the best-fit line are linear when plotted in this way.



17.17 Given these data

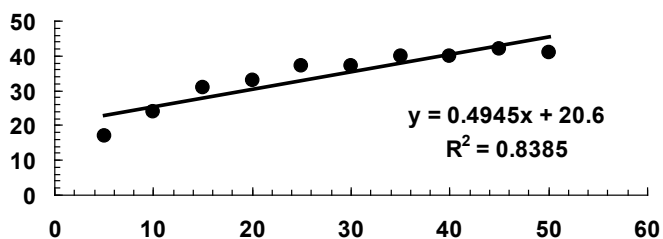
x	5	10	15	20	25	30	35	40	45	50
y	17	24	31	33	37	37	40	40	42	41

use least-squares regression to fit **(a)** a straight line, **(b)** a power equation, **(c)** a saturation-growth-rate equation, and **(d)** a parabola. Plot the data along with all the curves. Is any one of the curves superior? If so, justify.

(a) We regress y versus x to give

$$y = 20.6 + 0.494545x$$

The model and the data can be plotted as



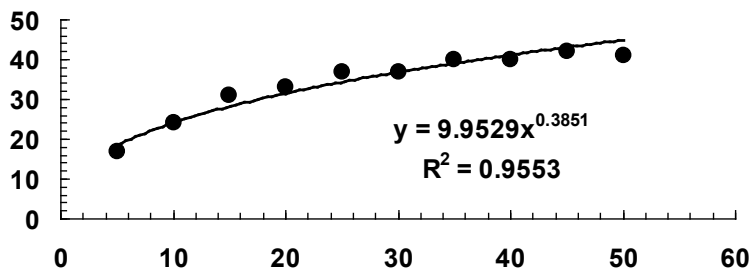
(b) We regress $\log_{10}y$ versus $\log_{10}x$ to give

$$\log_{10} y = 0.99795 + 0.385077 \log_{10} x$$

Therefore, $\alpha_2 = 10^{0.99795} = 9.952915$ and $\beta_2 = 0.385077$, and the power model is

$$y = 9.952915x^{0.385077}$$

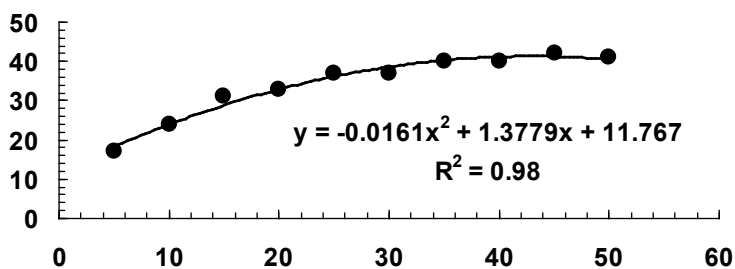
The model and the data can be plotted as



(d) We employ polynomial regression to fit a parabola

$$y = -0.01606x^2 + 1.377879x + 11.76667$$

The model and the data can be plotted as



Comparison of fits: The linear fit is obviously inadequate. Although the power fit follows the general trend of the data, it is also inadequate because (1) the residuals do not appear to be randomly distributed around the best fit line and (2) it has a lower r^2 than the saturation and parabolic models.

The best fits are for the saturation-growth-rate and the parabolic models. They both have randomly distributed residuals and they have similar high coefficients of determination. The saturation model has a slightly higher r^2 . Although the difference is probably not statistically significant, in the absence of additional information, we can conclude that the saturation model represents the best fit.

18.3 Estimate the natural logarithm of 10 using linear interpolation.

(a) Interpolate between $\ln 8 = 2.0794415$ and $\ln 12 = 2.4849066$.

(b) Interpolate between $\ln 9 = 2.1972246$ and $\ln 11 = 2.3978953$. For each of the interpolations, compute the percent relative error based on the true value.

(a)

$$f_1(10) = 2.079442 + \frac{2.484907 - 2.079442}{12 - 8}(10 - 8) = 2.282174$$

$$\varepsilon_t = \frac{2.302585 - 2.282174}{2.302585} \times 100\% = 0.886\%$$

(b)

$$f_1(10) = 2.197225 + \frac{2.397895 - 2.197225}{11 - 9}(10 - 9) = 2.29756$$

$$\varepsilon_t = \frac{2.302585 - 2.29756}{2.302585} \times 100\% = 0.218\%$$

Question 18.4

18.4 INTERPOLATE $\ln(10)$ From

$f(x_0) \ln 8 = 2.0794415$	$x_0 = 8$
$f(x_1) \ln 9 = 2.1972246$	$x_1 = 9$
$f(x_2) \ln 11 = 2.3978953$	$x_2 = 11$

USING 2ND ORDER POLYNOMIAL.

Eqn 18.4 $b_0 = 2.0794415$

18.5 $b_1 = \frac{2.1972246 - 2.0794415}{9 - 8}$
 $= 0.1177831$

18.6 $b_2 = -5.8157 \times 10^{-3}$

Then

$f_2(10) = 2.303375$

Textbook solution...

18.4 Fit a second-order Newton's interpolating polynomial to estimate $\ln 10$ using the data from Prob. 18.3 at $x = 8, 9$, and 11 . Compute the true percent relative error.

First, order the points

$$x_0 = 9 \quad f(x_0) = 2.197224577$$

$$x_1 = 11 \quad f(x_1) = 2.397895273$$

$$x_2 = 8 \quad f(x_2) = 2.079441542$$

Applying Eq. (18.4)

$$b_0 = 0.9542425$$

Equation (18.5) yields

$$b_1 = \frac{2.397895 - 2.197225}{11 - 9} = -0.00582$$

Equation (18.6) gives

$$b_2 = \frac{\frac{2.079442 - 2.397895}{8 - 11} - 0.100335}{8 - 9} = \frac{0.106151 - 0.100335}{8 - 9} = -0.00582$$

Substituting these values into Eq. (18.3) yields the quadratic formula

$$f_2(x) = 2.197225 + 0.100335(x - 9) - 0.00582(x - 9)(x - 11)$$

which can be evaluated at $x = 10$ for

$$f_2(x) = 2.197225 + 0.100335(10 - 9) - 0.00582(10 - 9)(10 - 11) = 2.303376$$

as cross-check, we can calculate the true relative error, and it should be less than the first order:

$$\varepsilon_t(\%) = \left| \left(\frac{2.303376 - 2.302585}{2.302585} \right) \right| \times 100\% = 0.034341\%$$

18.5 Fit a third-order Newton's interpolating polynomial to estimate $\ln 10$ using the data from Prob. 18.3.

Problem 18.5 (see also example 18.3 in the textbook)

Data	x	f(x)=ln(x)
x0	8	2.079442
x1	9	2.197225
x2	11	2.397895
x3	12	2.484907

General form:

$$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$$

or

$$f_3(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$

$$b_0 = f(x_0)$$

$$b_0 = 2.079442$$

$$f[x_1, x_0] = b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_1 = 0.117783$$

$$f[x_2, x_1] = 0.100335$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

$$b_2 = -0.00582$$

$$f[x_3, x_2] = 0.087011$$

$$f[x_3, x_2, x_1, x_0] = 0.000344$$

$$b_3 = 0.000344$$

$$f[x_2, x_1, x_0] = -0.00582$$

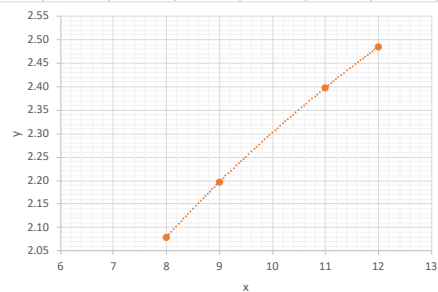
$$f[x_3, x_2, x_1] = -0.00444$$

$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

x	f3(x)	ln(10)
10	2.302689	2.302585

et 0.004495 %



Or using divided differences:

*** Note, the true error calculated in the textbook solution below is not quite right. it should be the $\text{abs}\{\ln(10) - 2.302689\} / \ln(10)$. The correct answer is shown above.

First, order the points

$$x_0 = 9 \quad f(x_0) = 2.197224577$$

$$x_1 = 11 \quad f(x_1) = 2.397895273$$

$$x_2 = 8 \quad f(x_2) = 2.079441542$$

$$x_3 = 12 \quad f(x_3) = 2.48490665$$

The first divided differences can be computed as

$$f[x_1, x_0] = \frac{2.397895 - 2.197225}{11 - 9} = 0.100335$$

$$f[x_2, x_1] = \frac{2.079442 - 2.397895}{8 - 11} = 0.106151$$

$$f[x_3, x_2] = \frac{2.484907 - 2.079442}{12 - 8} = 0.101366$$

The second divided differences are

$$f[x_2, x_1, x_0] = \frac{0.106151 - 0.100335}{8 - 9} = -0.00582$$

$$f[x_3, x_2, x_1] = \frac{0.101366 - 0.106151}{12 - 11} = -0.00478$$

The third divided difference is

$$f[x_3, x_2, x_1, x_0] = \frac{-0.00478 - (-0.00582)}{12 - 9} = 0.000344$$

Substituting the appropriate values into Eq. (18.7) gives

$$f_3(x) = 2.197225 + 0.100335(x - 9) - 0.00582(x - 9)(x - 11) + 0.000344(x - 9)(x - 11)(x - 8)$$

which can be evaluated at $x = 10$ for

$$f_3(10) = 2.197225 + 0.100335(10 - 9) - 0.00582(10 - 9)(10 - 11) + 0.000344(10 - 9)(10 - 11)(10 - 8) = 2.302689$$

as cross-check, we can calculate the true relative error, and it should be less than the second order:

$$\varepsilon_t(\%) = \left| \left(\frac{2.303376 - 2.302689}{2.302585} \right) \right| \times 100\% = 0.004492\%$$

18.15 Develop quadratic splines for the first five data points in Prob. 18.7 and predict $f(3.4)$ and $f(2.2)$.

For the present problem, we have five data points and $n = 4$ intervals. Therefore, $3(4) = 12$ unknowns must be determined. Equations 18.29 and 18.30 yield $2(4) - 2 = 6$ conditions

$$4a_1 + 2b_1 + c_1 = 8$$

$$4a_2 + 2b_2 + c_2 = 8$$

$$6.25a_2 + 2.5b_2 + c_2 = 14$$

$$6.25a_3 + 2.5b_3 + c_3 = 14$$

$$10.24a_3 + 3.2b_3 + c_3 = 15$$

$$10.24a_4 + 3.2b_4 + c_4 = 15$$

Passing the first and last functions through the initial and final values adds 2 more

$$2.56a_1 + 1.6b_1 + c_1 = 2$$

$$16a_4 + 4b_4 + c_4 = 8$$

Continuity of derivatives creates an additional $4 - 1 = 3$.

$$4a_1 + b_1 = 4a_2 + b_2$$

$$5a_2 + b_2 = 5a_3 + b_3$$

$$6.4a_3 + b_3 = 6.4a_4 + b_4$$

Finally, Eq. 18.34 specifies that $a_1 = 0$. Thus, the problem reduces to solving 11 simultaneous equations for 11 unknown coefficients,

$$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6.25 & 2.5 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6.25 & 2.5 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 10.24 & 3.2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10.24 & 3.2 & 1 \\ 1.6 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 16 & 4 & 1 \\ 1 & 0 & -4 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 1 & 0 & -5 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6.4 & 1 & 0 & -6.4 & -1 & 0 \end{bmatrix} \begin{Bmatrix} b_1 \\ c_1 \\ a_2 \\ b_2 \\ c_2 \\ a_3 \\ b_3 \\ c_3 \\ a_4 \\ b_4 \\ c_4 \end{Bmatrix} = \begin{Bmatrix} 8 \\ 8 \\ 14 \\ 14 \\ 15 \\ 15 \\ 2 \\ 8 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

which can be solved for

$$b_1 = 15 \quad c_1 = -22$$

$$a_2 = -6 \quad b_2 = 39 \quad c_2 = -46$$

$$a_3 = -10.816327 \quad b_3 = 63.081633 \quad c_3 = -76.102041$$

$$a_4 = -3.258929 \quad b_4 = 14.714286 \quad c_4 = 1.285714$$

The predictions can be made as

$$f(3.4) = -3.258929(3.4)^2 + 14.714286(3.4) + 1.285714 = 13.64107$$

$$f(2.2) = -6(2.2)^2 + 39(2.2) - 46 = 10.76$$

Finally, here is a plot of the data along with the quadratic spline,

